The Oldest and The Youngest.

"In these days of conflict between ancient and modern studies; there must surely be something to be said for a study which did not begin with Pythagoras and will not end with Einstein; but is the oldest and the youngest." — G.H. Hardy

Set Theory: The fundamental language of modern mathematics. It defines relations, functions, geometry, sequences, and probability.
Origin: Developed by Georg Cantor (1845–1918).

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The Criterion of Definition

A set is a well-defined collection of objects.

Ambiguous Group

A crowd of people

Five most renowned mathematicians

NOT A SET. Criteria varies from person to person.

Mathematical Set

- The rivers of India
- The vowels in the English alphabet

SET. Membership is objective and definite.

Binary Rule of Membership

$$a \in A$$

(Element 'a' belongs to Set A)

$$b \notin A$$

(Element 'b' does not belong to Set A)

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Modes of Representation

Roster Form (The List)

$$\{1, 2, 3, 6, 7, 14, 21, 42\}$$

- **Method:** Listing elements within braces.
- **Rule 1:** Order is immaterial.
  $\{1, 2, 3\} = \{3, 1, 2\}$
- **Rule 2:** No repetition.
  "SCHOOL" → $\{S, C, H, O, L\}$

Set-Builder Form (The Rule)

$$\left\{ x : x \text{ is a natural number which divides } 42 \right\}$$

- **Method:** Defining the common property.
- **Syntax:** $\{ x : \text{property of } x \}$
- **Example:**
  $V = \{ x : x \text{ is a vowel in the English alphabet} \}$

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The Nature of Existence

The Empty Set

$$\{\varphi\}$$

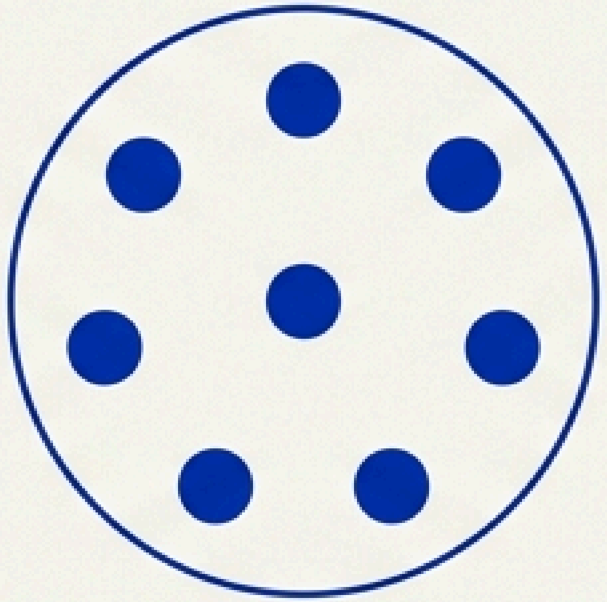
A set containing no elements.

Example:

Impossible

$\{ x : x \text{ is a natural number}, 1 < x < 2 \}$

Finite Sets

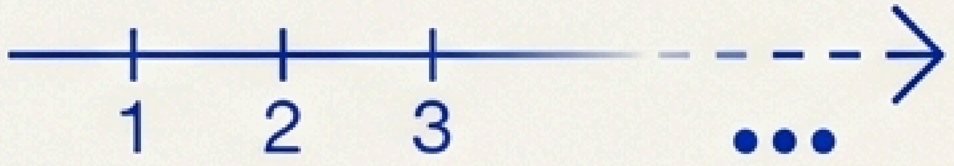


A set with a definite number of elements.

Example:

Days of the week (n=7)

Infinite Sets



A set where the number of elements is not finite.

Example:

Natural Numbers $\mathbb{N} = \{1, 2, 3...\}$

Infinite sets like Real Numbers ($\mathbb{R}$) cannot be written in roster form as they lack a simple pattern.

Equality

Two sets A and B are equal (A = B) if they have exactly the same elements.

$$\boxed{\{1, 2, 3, 4\} = \{3, 1, 4, 2\}}$$

Identity exists irrespective of order.

Word: ALLOY → Set
$X = \{A, L, L, O, Y\}$

$$X = B$$

Word: LOYAL → Set
$B = \{L, O, Y, A, L\}$

Repetition does not change the set.

Inequality: If even one element differs (e.g., $0 \in A$ but $0 \notin B$), then $A \neq B$.

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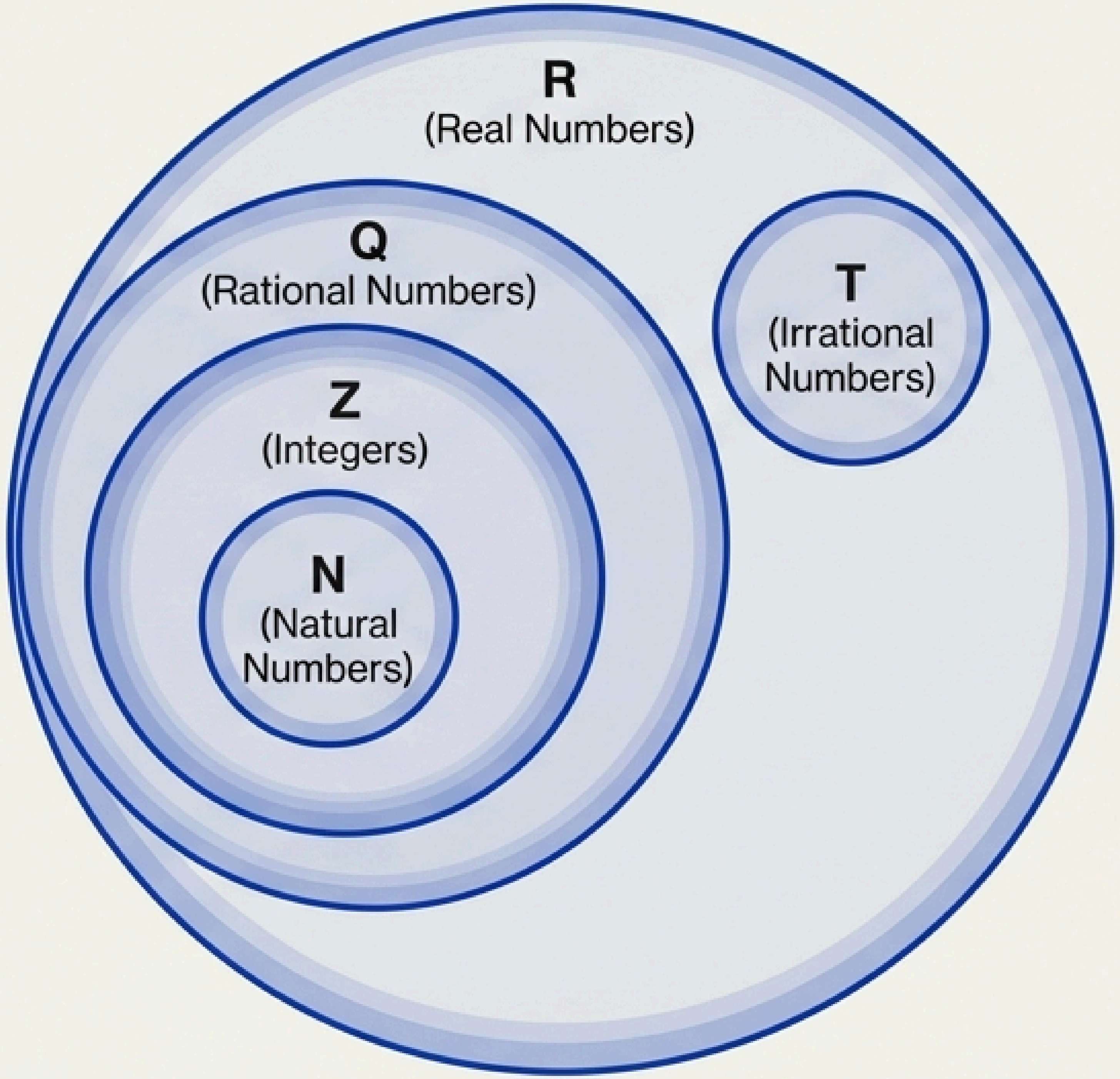
Subsets and Supersets

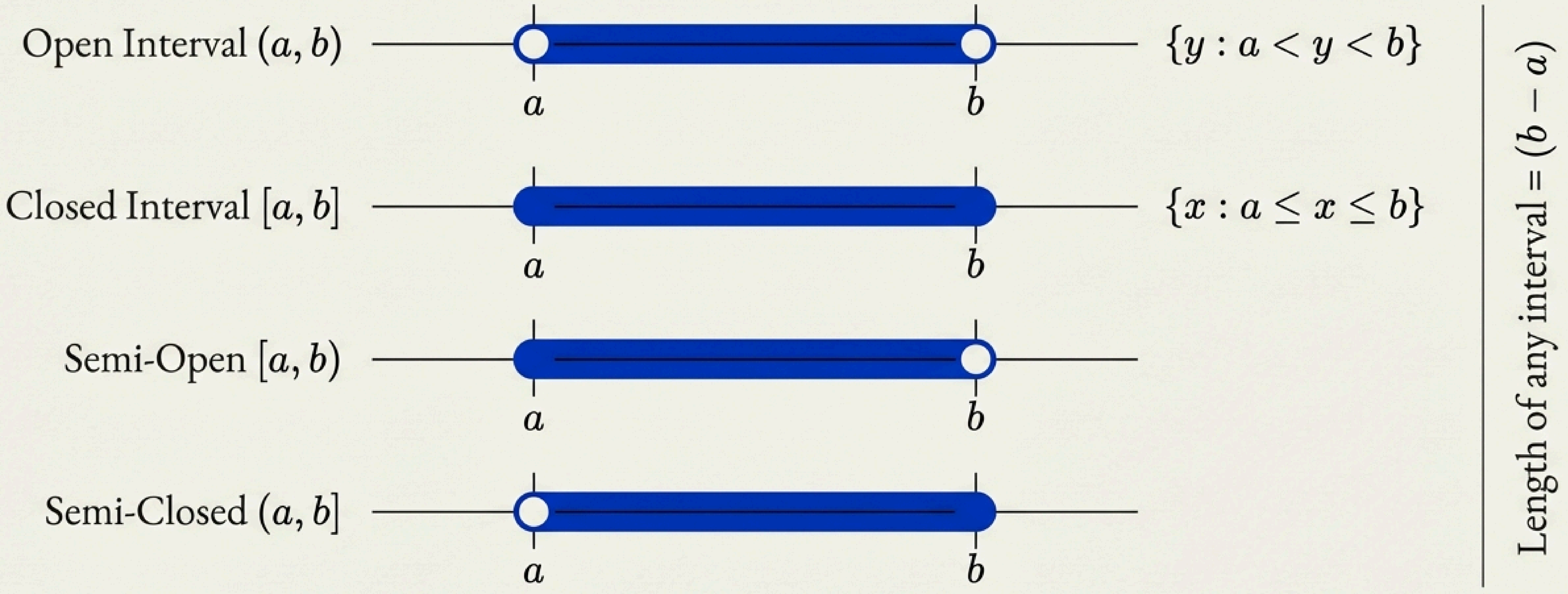
$A \subset B$ if every element of A is also an element of B.



Key Rules:

1. Every set is a subset of itself ($A \subset A$).
2. The empty set is a subset of every set ($\varphi \subset A$).
3. If $A \subset B$ and $A \neq B$, then A is a proper subset.

Intervals as Subsets of Real Numbers



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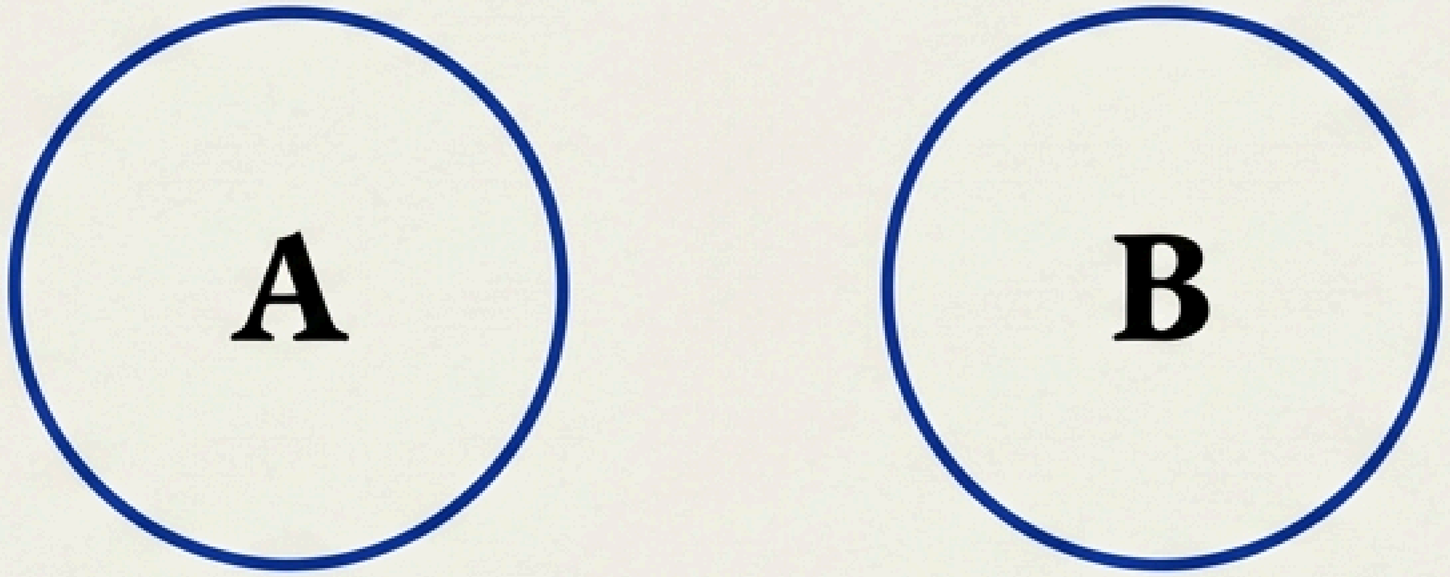
The Universal Set (U)

U

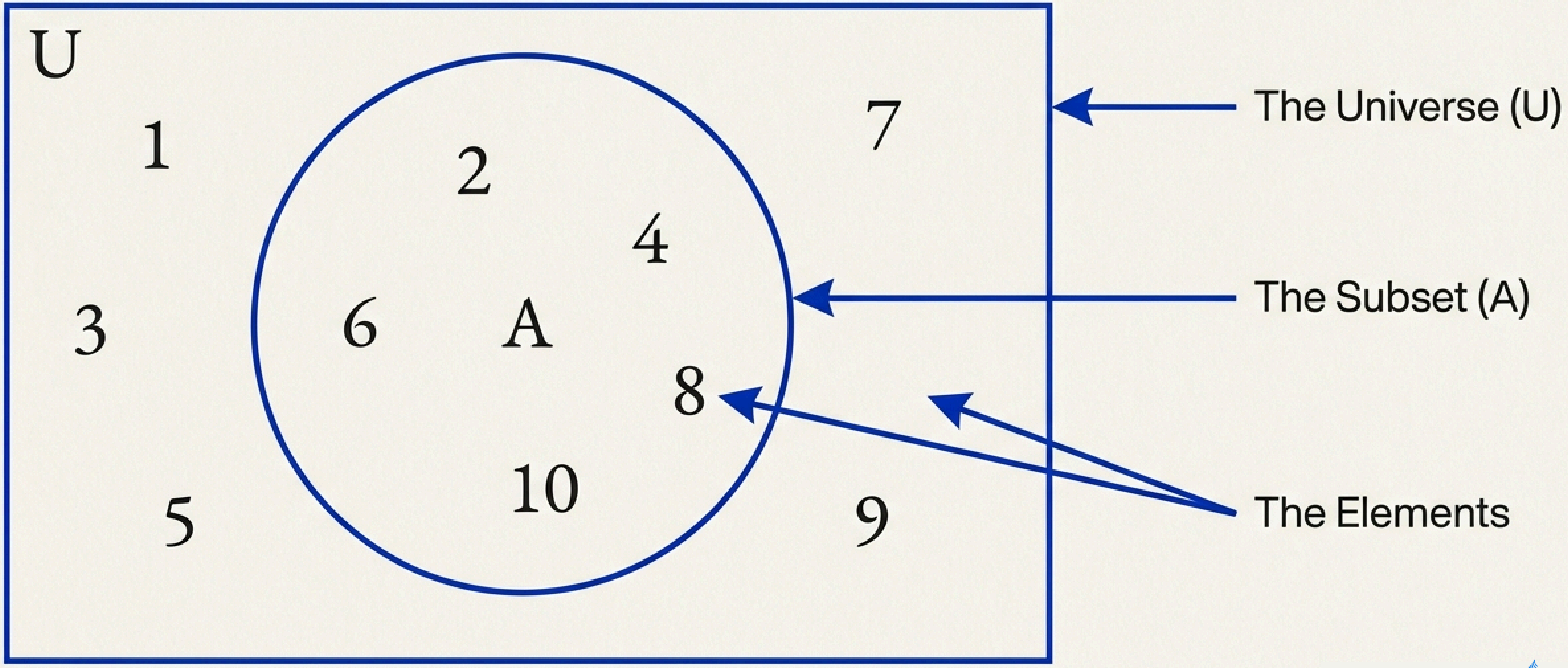
Concept: The "Canvas"—the basic set relevant to a particular context containing all other subsets.



- Context: Integers. Universal Set: Rational Numbers (Q).
- Context: Population Studies. Universal Set: All people in the world.
- Context: Right Triangles. Universal Set: All triangles.

The Visual Lexicon: Venn Diagrams

Mapping logic to geometry. Named after John Venn (1834–1883).



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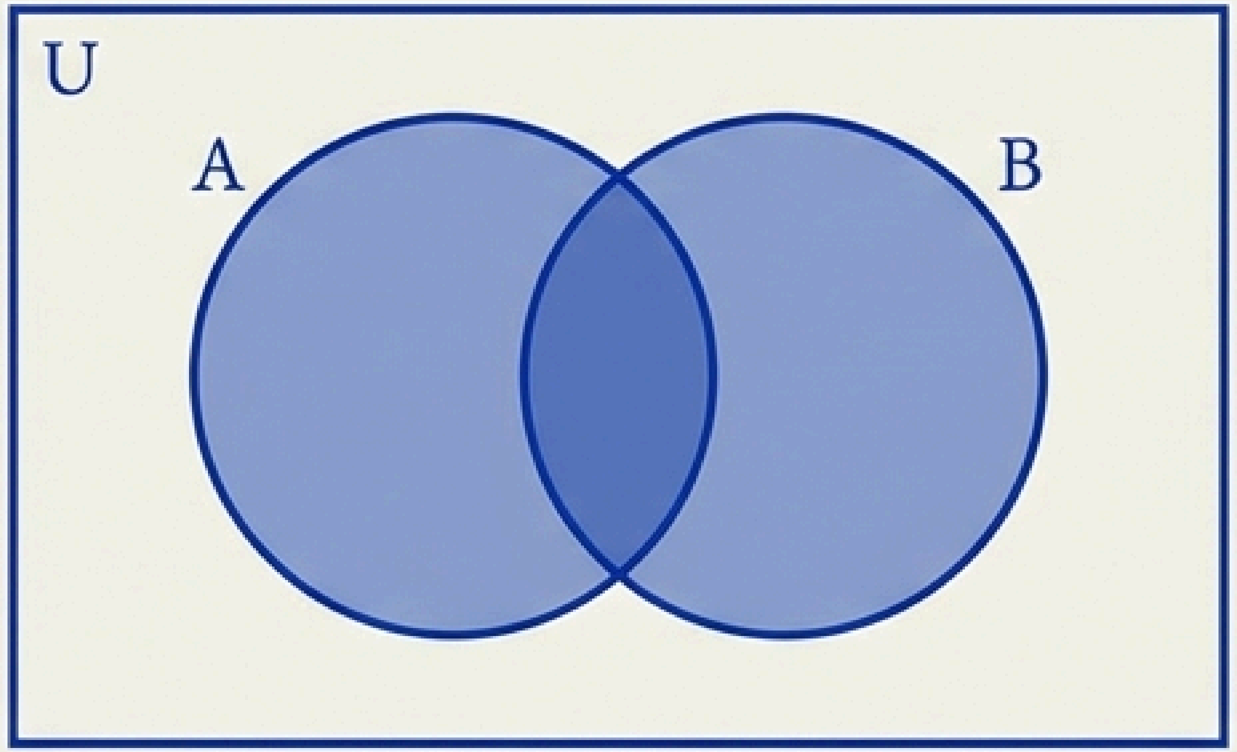
Operation: Union (∪)

The set of elements which are in A, OR in B, OR in both.

$$A \cup B = \{ x : x \in A \text{ or } x \in B \}$$



A = { Ram, Geeta, Akbar }

Y = { Geeta, David, Ashok }

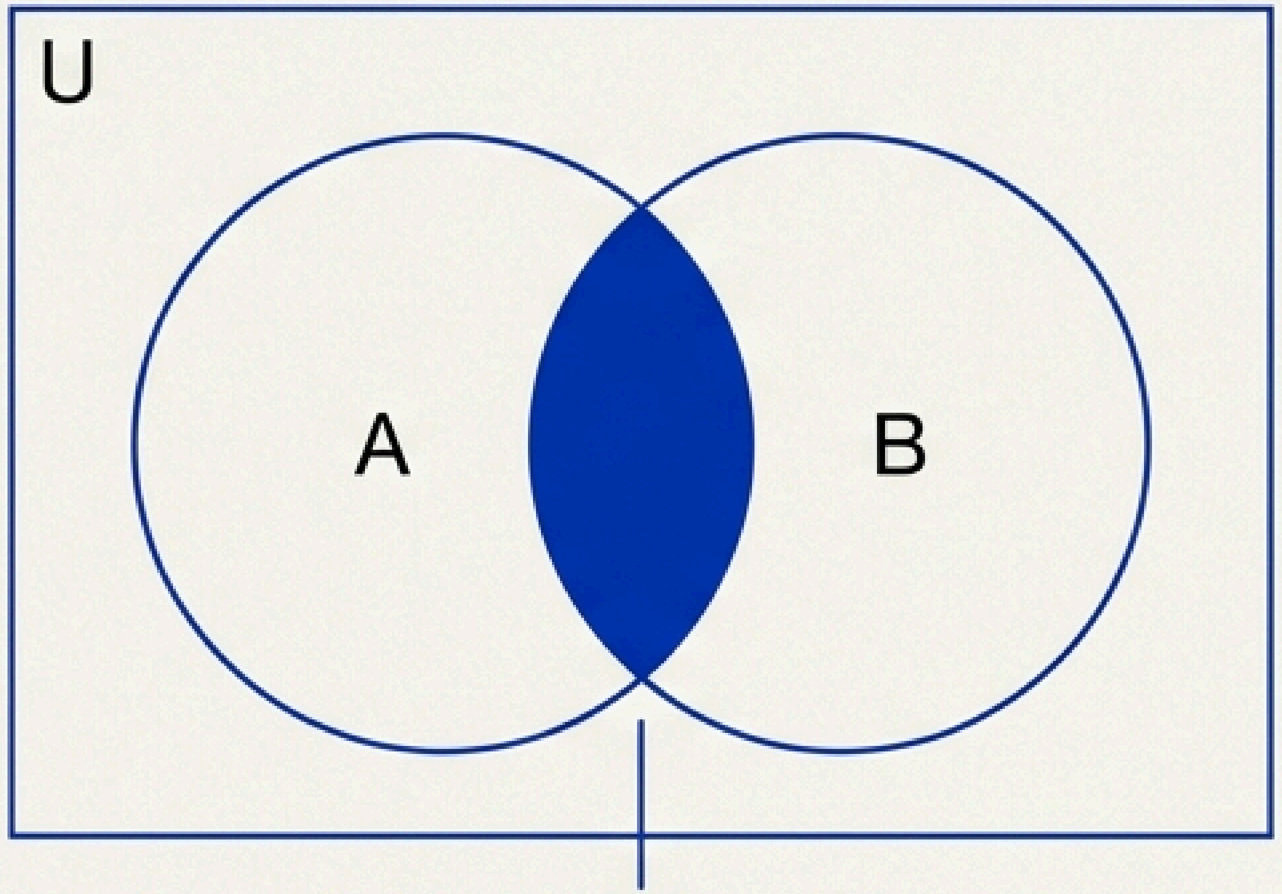
A ∪ Y = { Ram, Geeta, Akbar, David, Ashok }

Common element 'Geeta' is listed only once.

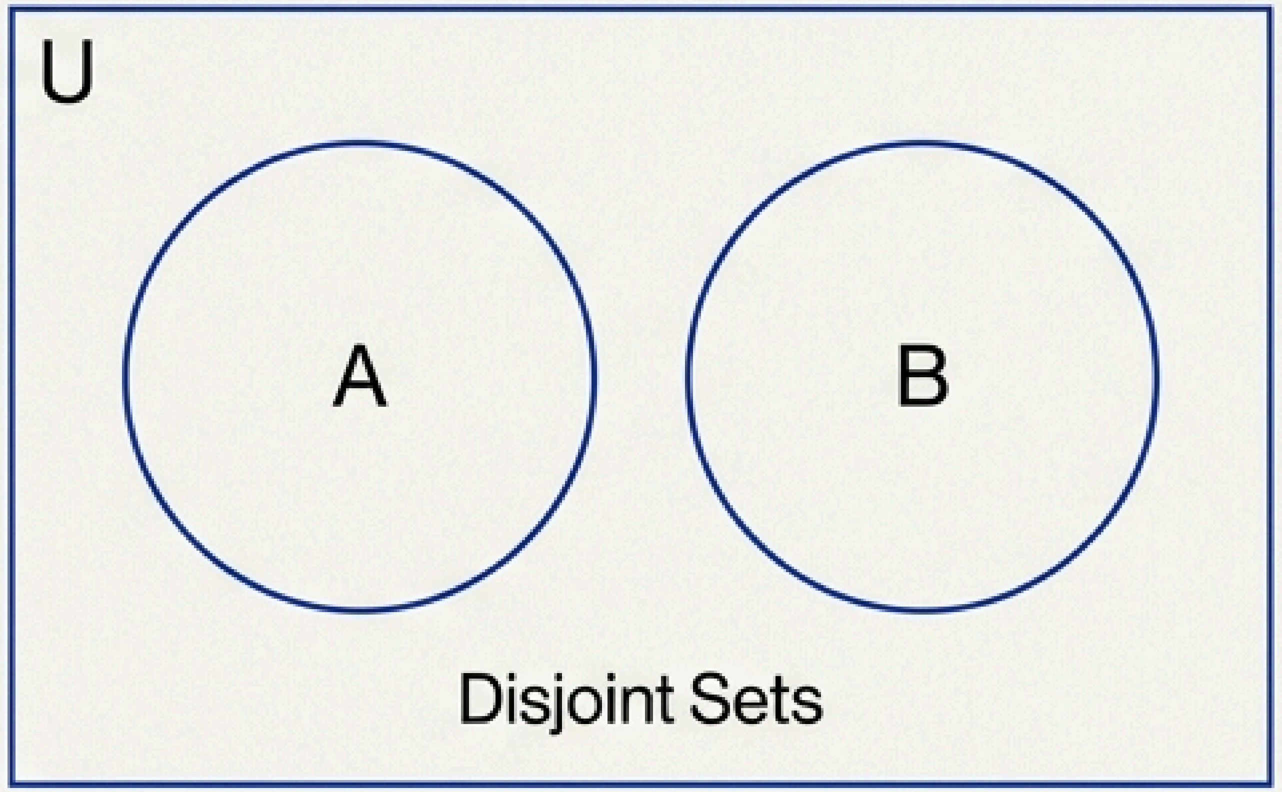
Operation: Intersection (∩)

The set of elements common to BOTH A and B.

$$A \cap B = \{ x : x \in A \text{ and } x \in B \}$$



$A \cap B$



$A \cap B = \phi$ (No common ground)

$\{2, 4, 6\}$ and $\{1, 3, 5\}$

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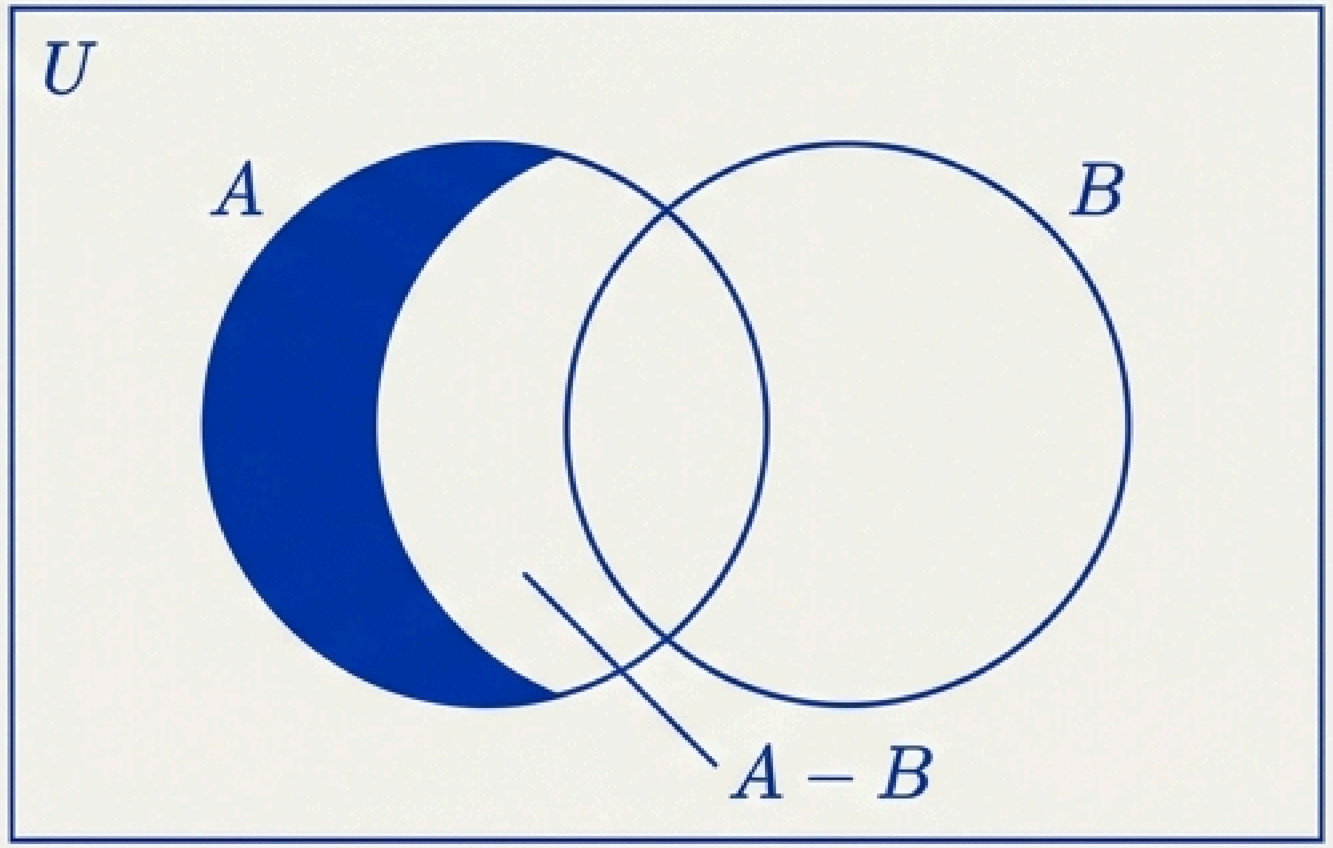
Operation: Difference (-)

The set of elements belonging to A but NOT to B.

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$



Critical Insight
- Order Matters: A - B $\neq$ B - A
- The sets (A - B), (A $\cap$ B), and (B - A) are mutually disjoint.

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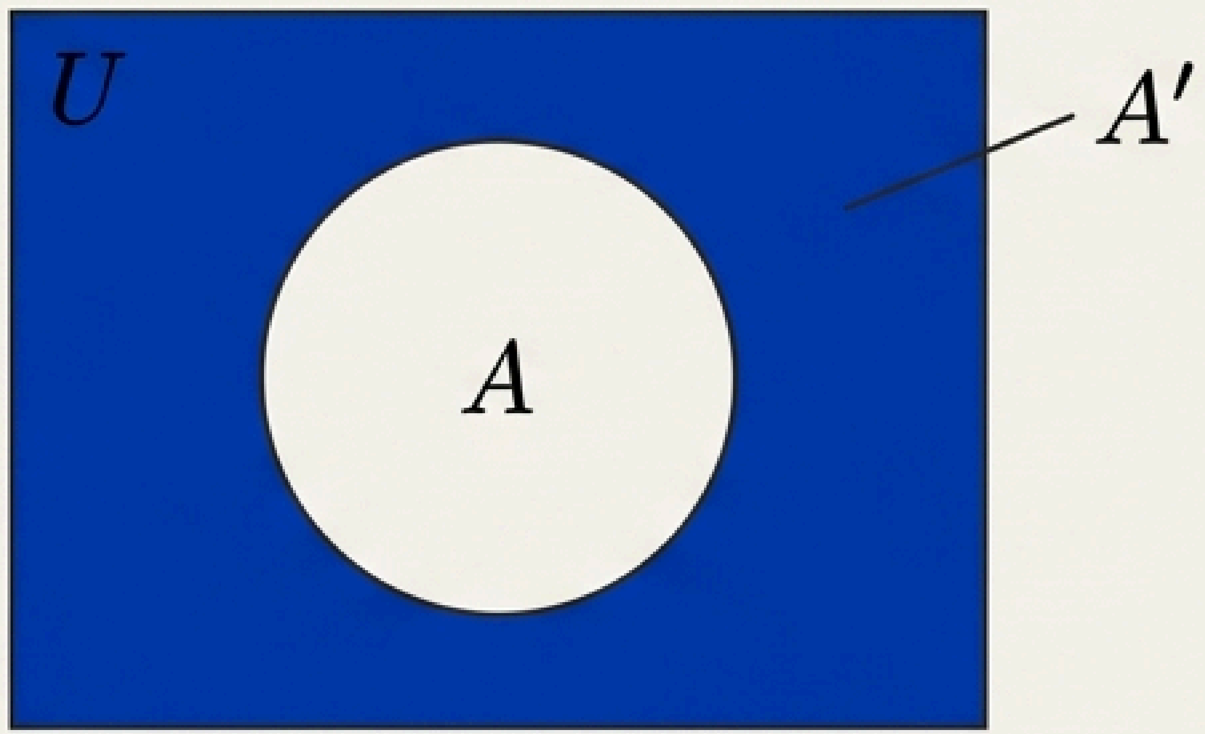
Operation: Complement (’)

The set of all elements in the Universe (U) that are NOT in A.

$$A' = U - A = \{\, x : x \in U \text{ and } x \notin A \,\}$$



Double Complementation Law: (A’)’ = A

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The Laws of Set Theory

Cheat Sheet

1	**De Morgan's Laws**	
	$(A \cup B)' = A' \cap B'$	(Complement of union is intersection of complements)
	$(A \cap B)' = A' \cup B'$	(Complement of intersection is union of complements)
2	**Fundamental Laws**	
	Idempotent: $A \cup A = A \ ; \ A \cap A = A$	(Garamond m idempttent)
	Identity: $A \cup \phi = A \ ; \ A \cap U = A$	(Garamond is identity)
	Complement: $A \cup A' = U \ ; \ A \cap A' = \phi$	(Garamond in complement)

A Legacy of Logic

History:

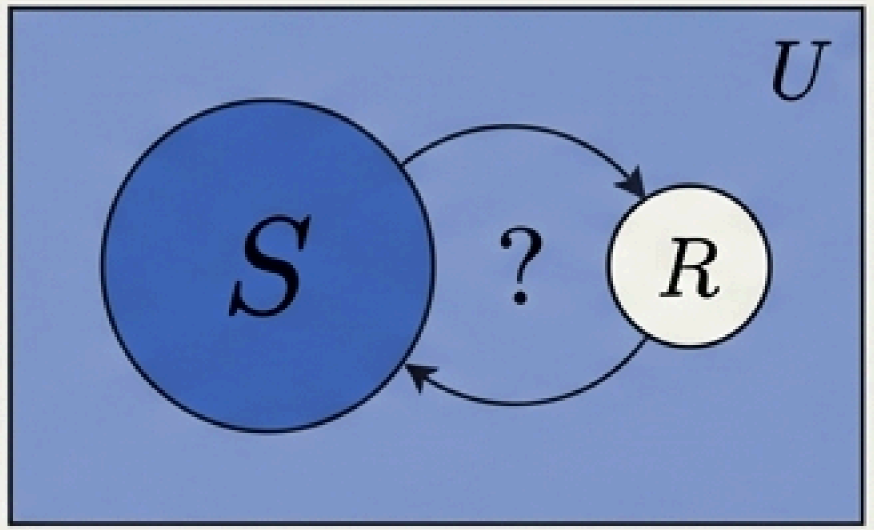
Georg Cantor's work on infinite sets faced initial resistance (e.g., Kronecker: "God made the integers, all else is the work of man").

Refinement:

Formalized by Zermelo and Fraenkel to resolve logical cracks.

Russell's Paradox (1902)

Assuming a "set of all sets" exists leads to contradiction.
Conclusion: Nothing contains everything.



$$R = \{\, x : x \notin x \}$$

Despite the paradoxes, Set Theory remains the operating system of modern mathematics.